

UTS Cloud Computing Class - Lab Guide

Week 2 Lab: VMware ESXi, Build VMs

In this lab, you will be ensuring that your environment is functioning as expected, and then interacting with several of the virtual machines. ***Note:** Not all virtual machines need to be accessed to complete this lab.

Task 1. Verify Access

Ensure all files are accessible.

1. Connect to the network share (/class/CloudComputing/master) and copy ESXi01 and ESXi02 into the folder called (/images/tmp/<student ID>). Create a folder with your student ID under /images/tmp/.
2. Open VMware Workstation and attach the VMs to VMware Workstation inventory by selecting **File > Open** and browsing to the .vmx location of each one of your VMs.
3. Verify that you have 2 VM in the inventory of your VMware Workstation software. Their names should be: 'ESXi01' and 'ESXi02'.
4. Power ON two of the VMs: ESXi01 and ESXi02. Select the VM in the left-hand inventory panel, then click the '**Power on this Virtual Machine**' link in the main window.

Note:

All the files can also be downloaded from GoogleDrive through the link:

<https://drive.google.com/drive/folders/1ljiPLqRPCpnKLaoPqsuESWflPXV6mkxr?usp=sharing>

Task 2. Verify VMs

Ensure VMs can be accessed.

*All passwords unless otherwise stated are: '**VMware1!**'*

1. Login to ESXi01 using the **root** user and above given password. Click on the window and press F2 to grab control.
2. Select "Configure Management Network" to check the IP of the host. For ESXi01, the IP is set to 172.20.20.51. Gateway 172.20.20.x
3. Similarly, repeat the step for ESXi02. The IP for ESXi02 is set to 172.20.20.52. Gateway 172.20.20.1
4. Right click ESXi 1 and 2, choose network adaptor, select custom, choose vmnet6

Task 3. Create a new ESXi host (as a VM).

From VMware Workstation, create a new virtual machine 'Esxi03'. Assign the following parameters:

- ☐ 2 cpus
- ☐ 4GB ram
- ☐ 18GB disk
- ☐ 4 NICs in VMnet mode

1. File->New Virtual Machine. In the Wizard, select **Typical**, then set the new virtual machine according to requirements.

2. Locate the ESXi install iso (xxxinstallerxxx.iso) on the class cloud share and attach it to the CD drive of the newly created VM.
3. Create ESXi in the location of /images/tmp/<student ID>
4. Select "Customize Hardware" to add more NICs.
5. Power up the VM, set password '**VMware1!**' for the user root and complete a local install.
6. Once installed, access the console by press **F2** and configure the IP address of the management network.
7. Login as the **root** user
8. Select "Configure Management Network" and "IP Configuration" 9. Choose "static IP"
172.20.20.53/24 Gateway 172.20.20.1
10. Select "Restart Management Network".

Note: To configure IP address

1. Login to the ESXi host as the **root** user
2. Select "Network Restore Options", then "Restore Network Settings".
3. You might want to "Restart Management Network". After that, you should be able to see the IP changed.
4. Select "Configure Management Network" and "IP Configuration" 7. Choose "static IP"
172.20.20.51(52)(53)/24 Gateway 172.20.20.1
8. Select "Restart Management Network".

Till here, you have finish most of the tasks for this week. If you still have lots of time, you can start working on Task 4 and Task 5, otherwise jump to Task 6.

Task 4 Manage ESXi hosts To manage the ESXi VMs, the host computer's web browser interface will be used.

1. Open Mozilla Firefox web browser and type the URL for respective ESXi host. For e.g.
<https://172.20.20.51/> for ESXi01.
2. VMware login page should open. The credentials to login are the same as used in Task 2. 3. Similarly, type the IP for ESXi02 to manage through web interface.

Note: To assign a license, click Manage->Licensing->Assign license, then input the key:

JJ2WR-25L9P-H71A8-6J20P-C0K3F

Task 5 Create New Virtual Machines

After the login to each ESXi host, the page will give the summary of the ESXi host.

1. In the left panel select the '**Virtual Machines**' option.
2. Select '**Create / Register VM**' to create a new virtual machine within the ESXi01 host.

3. Follow the steps and enter following values:
 - Virtual Machine Name = *YourName-vm1*
 - Guest OS family = Linux
 - Guest OS version = Ubuntu Linux (64-bit)
 - Memory = 384MB
 - Hard Disk = 1GB - Thin Provisioned *Leave all the other values as defaults.*
 4. Once the VM has been created, select the VM and select '**Power On**'
 5. Right-Click the VM again and select '**Console -> Open browser console**'
 6. Is there an error message at the console of the New Machine that you created? Explain what the problem is: _____
 7. Power Off vm1
 8. Repeat steps 2 - 4 from for ESXi02 using the suffix -vm2 for your second VM.
 9. Right-Click vm1 and select '**Edit Settings**'
 10. Select the **CD/DVD Drive1**, locate the **TinyCore-current.iso** which is on the local datastore of the ESXi host that the vm is running from in the **ISO** folder.
 11. Select '**Connect**' checkbox then select '**Save**'.
 12. Select the Console of vm1 again, and now click the Power On link at the top menu bar. The VM will now boot from CD drive where the OS iso file is attached.
 13. Verify that an unattended TinyCore linux install is initiated.
 14. Repeat Steps 10 - 13 for vm2 on ESXi02.
- You should now have created 2 VMs, each on a different host.

Task 6. Tidy Up

We now need to shut down the lab until next week.

1. Navigate to any running VMs of ESXi01, and ESXi02.
2. Right-Click them and Select '**Shutdown Guest**'.
3. Once complete, select the '**Shutdown**' option for ESXi01 and ESXi02.
4. This will disconnect your ESXi host connection.
5. Return to VMware Workstation and verify that ESXi01 and ESXi02 are properly shut down.
6. Close VMware Workstation.